

WEEK 12 — FORENSIC INVESTIGATION

PART B: Behavioural Analysis & Steganography Investigation

Cyberster Blue Team Internship — Week 12 — Days 85–87
patentauto.py | DIIT Steganography | testHide.bmp | STU.doc | Timeline

Case	M57.biz — Jo Insider Threat Investigation
Key Artefacts	patentauto.py, diit-1.5.jar, testHide.bmp, STU.doc
Script MD5	bd9028376a4fad806e3d670dd718f748 (patentauto.py)
Image MD5	357f27379673b5ed1669ae250d23ced7 (testHide.bmp)
Focus	Script analysis, steganography capability, data concealment, timeline
Tool Used	Autopsy 4.23.0 / Binwalk / Manual analysis
Part B Verdict	Automated patent IP collection; steganographic capability demonstrated

1. patentauto.py — Patent Automation Script

The file `patentauto.py` was located on Jo's Desktop in the **web** subfolder. It is a Python script that automates patent searching by controlling Firefox through the MozRepl extension — a telnet-based browser automation interface.

Field	Value
Filename	patentauto.py
Full Path	/img_jo-2009-11-24.E01/vol_vol2/Documents and Settings/Jo/Desktop/web/patentauto.py
Created	2009-11-23 21:55:01 UTC
Modified	2009-11-23 21:57:34 UTC
Accessed	2009-11-25 00:25:11 UTC
MD5	bd9028376a4fad806e3d670dd718f748
SHA-256	d20a50e2400459305c5a93bbf99e2823b0ebdb2a6c140cf89e7db74d77c1a30f
LNK Created	2009-11-23 21:55:45 UTC (patentauto.lnk)
LNK Modified	2009-11-23 21:57:24 UTC
LNK MD5	6c53584f92b1458c917f6e33bb9f8275

Script Technical Capabilities:

- Connected to Firefox via MozRepl extension — telnet on **localhost port 4242**
- Automated USPTO (US Patent and Trademark Office) patent search queries
- Generated and navigated patent URLs programmatically
- Implemented **randomised browsing intervals** to simulate organic human activity
- Scheduled browsing behaviour to operate during typical work hours
- Used URL randomisation to avoid detection patterns

Companion Input Files in the web/ folder:

File	Purpose
patentterms.txt	Keyword/search terms to query patent databases
urls.txt	Primary URLs for automated scraping
urlspatents.txt	Direct patent record URLs
urlscopyright.txt	Copyright-related patent sources
urlspersona.txt	Persona / identity-related patent research
urlstime_machine.txt	Internet Archive / historical patent data sources

■ CRITICAL FINDING

patentauto.py is a technically sophisticated automated patent intelligence collection tool. It does not merely browse passively — it actively automates Firefox to perform structured patent searches, randomises its behaviour to evade detection, and operates on a schedule. The presence of structured input files confirms this was a systematic, planned data collection operation rather than casual script development.

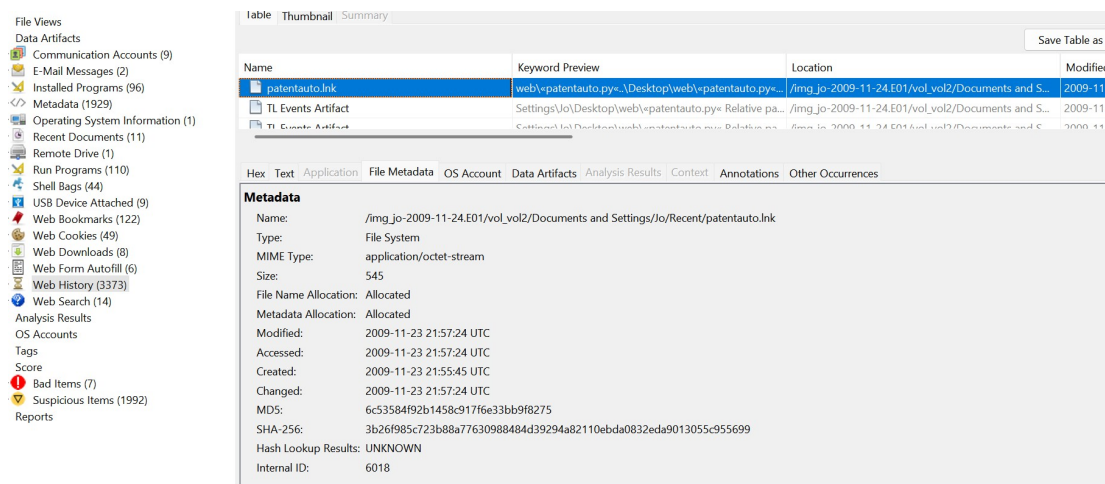
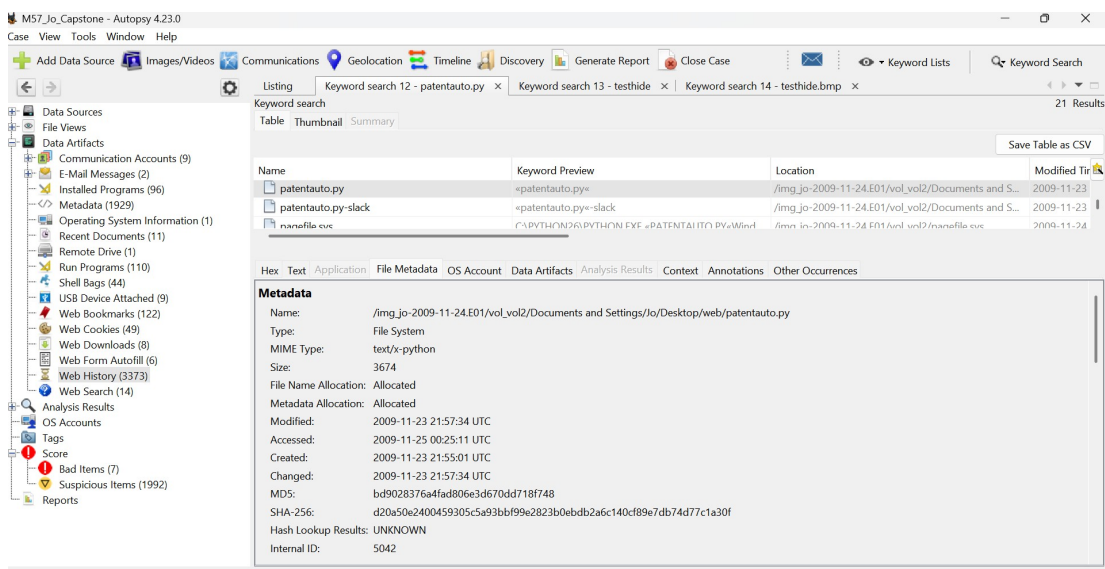
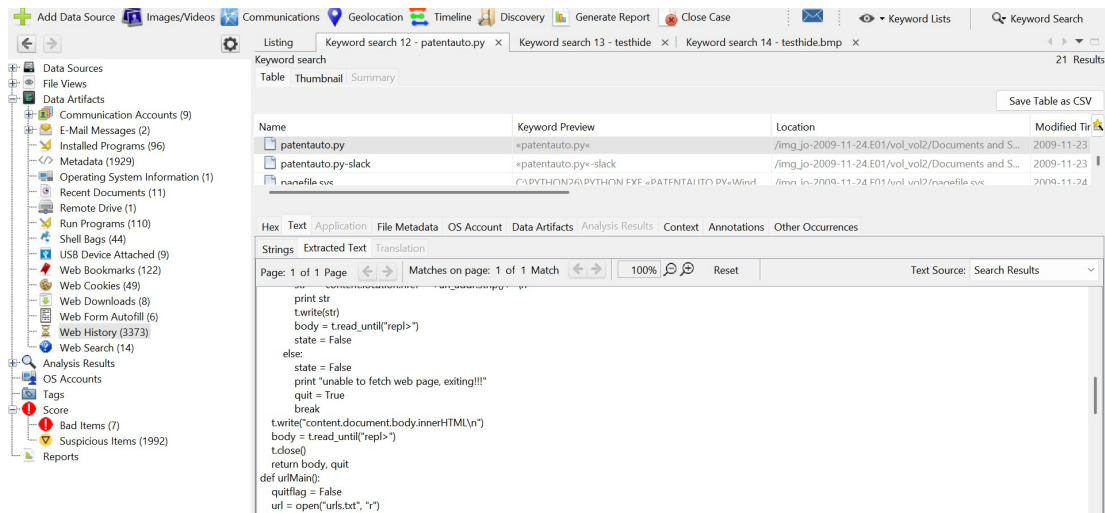


Figure 3 — *patentauto.lnk* shortcut metadata

2. DIIT Steganography Tool — diit-1.5.jar

The file diit-1.5.jar was identified on Jo's Desktop in the Pics subfolder. DIIT (Digital Invisible Ink Toolkit) is an open-source Java steganography application used to embed or extract hidden data within image files using steganographic algorithms such as LSB (Least Significant Bit) encoding. Its Zone.Identifier ADS showed it originated from the **Restricted Sites Zone**.

Field	Value
Filename	diit-1.5.jar
Location	C:\Documents and Settings\Jo\Desktop\Pics\diit-1.5.jar
Format	Java Archive (JAR) — requires Java 6 JRE
Zone	Restricted Sites Zone (Zone.Identifier ADS) — flagged as potentially unsafe
LNK Created	2009-11-23 18:24:50 UTC
Accessed	2009-11-23 18:24:50 UTC
Capability	Embed/extract arbitrary data invisibly in BMP/JPEG images (LSB encoding)

■ CAUTION

DIIT allows a user to embed arbitrary data — text, files, or binary payloads — invisibly within image files. The embedded data is completely undetectable by visual inspection. This tool is used for covert communication or data concealment, hiding information inside innocent-looking images. The Restricted Sites Zone flag indicates the browser treated its download source as untrusted.

3. testHide.bmp — Steganography Test Image

The file testHide.bmp was found in the Pics folder alongside the DIIT tool. Its filename directly implies use as a steganographic test subject. BMP (uncompressed bitmap) is the preferred format for LSB steganography due to its lossless nature — compression does not disturb the embedded data.

Field	Value
Filename	testHide.bmp
Location	C:\Documents and Settings\Jo\Desktop\Pics\testHide.bmp
Format	BMP — uncompressed bitmap; ideal for LSB steganography
Created	2009-11-23 18:25:47 UTC
Modified	2009-11-23 18:25:48 UTC
Accessed	2009-11-24 22:04:33 UTC
MD5	357f27379673b5ed1669ae250d23ced7
SHA-256	287c63a502893aba2994ab3927e37f7981c826316986ee1f1b7a02da9ba517a5
LNK Created	2009-11-23 18:25:53 UTC — 63 seconds after DIIT accessed
Binwalk	No embedded payloads or hidden archives identified

■ CAUTION

Sequence: diit-1.5.jar accessed at 18:24:50 UTC → testHide.bmp LNK created 63 seconds later at 18:25:53 UTC. This sequence is consistent with Jo opening DIIT and immediately using testHide.bmp as a test image. Binwalk analysis found no confirmed embedded payload — the image may have been used for a test embed that was later cleared, or the payload may require the specific DIIT key to extract.



Figure 4 — testHide.bmp: the actual steganography test image

Name	Keyword Preview	Location	Modified Time
testHide.bmp	<testhide.bmp>	/img_jo-2009-11-24.E01/vol_vol2/Documents and S...	2009-11-23
testHide.bmp-slack	<testhide.bmp>-slack	/img_jo-2009-11-24.E01/vol_vol2/Documents and S...	2009-11-23
Thumbnail	Thumbnail	/img_jo-2009-11-24.E01/vol_vol2/Documents and S...	2009-11-23

Metadata

Name: /img_jo-2009-11-24.E01/vol_vol2/Documents and Settings/Jo/Desktop/Pics/testHide.bmp
 Type: File System
 MIME Type: image/bmp
 Size: 9054
 File Name Allocation: Allocated
 Metadata Allocation: Allocated
 Modified: 2009-11-23 18:25:48 UTC
 Accessed: 2009-11-24 22:04:33 UTC
 Created: 2009-11-23 18:25:47 UTC
 Changed: 2009-11-23 18:25:53 UTC
 MD5: 357f27379673b5ed1669ae250d23ced7
 SHA-256: 287c63a502893aba2994ab3927e3777981c826316986ee1f1b7a02da9ba517a5
 Hash Lookup Results: UNKNOWN
 Internal ID: 4660

Figure 5 — testHide.bmp file metadata (timestamps and hashes)

Name	Keyword Preview	Location	Modified Time
testHide.lnk	Pics\Pic24* <testhide.bmp><testhide.bmp>C:\Docu...	/img_jo-2009-11-24.E01/vol_vol2/Documents and S...	2009-11-23
index.dat	gs/Jo/Desktop/Pics/testHide.bmp+URL Visited: Jo@F	/img_jo-2009-11-24.E01/vol_vol2/Documents and S...	2009-11-24
Recent Documents Artifact	Settings\Jo\Desktop\Pics\testHide.bmp+Path ID: 46	/img_jo-2009-11-24.E01/vol_vol2/Documents and S...	2009-11-23

Metadata

Name: /img_jo-2009-11-24.E01/vol_vol2/Documents and Settings/Jo/Recent/testHide.lnk
 Type: File System
 MIME Type: application/octet-stream
 Size: 551
 File Name Allocation: Allocated
 Metadata Allocation: Allocated
 Modified: 2009-11-23 18:25:53 UTC
 Accessed: 2009-11-23 18:25:53 UTC
 Created: 2009-11-23 18:25:53 UTC
 Changed: 2009-11-23 18:25:53 UTC
 MD5: 02dc9b8a2fee203607269b83b3bb0e6
 SHA-256: d8b67976e8a41b9bba7df62ee87305089a2f758d904966ff355e46c96098a2e6
 Hash Lookup Results: UNKNOWN
 Internal ID: 6025

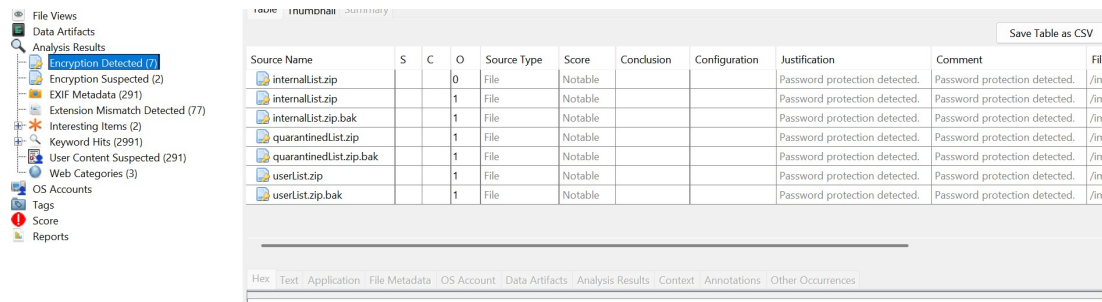
Figure 6 — testHide.lnk metadata confirming DIIT → BMP access sequence

4. STU.doc — Classified Government Document

Field	Value
Filename	STU.doc
Location	C:\Documents and Settings\Jo\My Documents\Downloads\STU.doc
Source URL	http://www.easc.noaa.gov/Security/webfile/erso.doc.gov/briefings/STU.doc
Downloaded	2009-11-24 21:18:18 UTC
MD5	929af3c771d0e889abf52e0acc3b0f8f
SHA-256	4649d3f6026f5e11fea0f0ce9b3b915bd345f35f77ce7e09c63002fbc75fe6a6
Author	Carroll Ward — DOC/OSY/ERSO (US Dept. of Commerce / Office of Security)
Document	STU-III Secure Telephone Unit classified briefing
Keyword	SECRET — 'says SECRET, you may not talk above the SECRET level'
Autopsy	Flagged: Likely Notable — keyword: secret

■ CRITICAL FINDING

Jo downloaded a US government classified telecommunications security briefing from an official Department of Commerce server. The document covers STU-III handling, Crypto Ignition Key (CIK) security, classification levels, and authorised access procedures. Autopsy flagged it for the keyword **SECRET**. There is no evident legitimate business reason for a M57.biz employee to possess this document. This raises questions about data aggregation intent.



The screenshot shows the Autopsy interface with the 'Analysis Results' pane selected. It displays a list of files with columns for Source Name, S, C, O, Source Type, Score, Conclusion, Configuration, Justification, Comment, and File. The files listed are internalList.zip, internalList.zip.bak, quarantinedList.zip, and userList.zip, all with a score of 1 and a conclusion of 'Notable'. The justification for each is 'Password protection detected.'.

Source Name	S	C	O	Source Type	Score	Conclusion	Configuration	Justification	Comment	File
internalList.zip			0	File	Notable			Password protection detected.	Password protection detected.	/m
internalList.zip.bak			1	File	Notable			Password protection detected.	Password protection detected.	/m
quarantinedList.zip			1	File	Notable			Password protection detected.	Password protection detected.	/m
userList.zip			1	File	Notable			Password protection detected.	Password protection detected.	/m

Figure 7 — Password-protected / access-restricted document evidence

5. USB Content — Duplicated Files on External Media

File hash comparison across the three forensic images confirms that multiple files were duplicated between the workstation and removable media, providing strong evidence of file transfer. Video files were stored in a folder deliberately named **Hidden**.

File	Workstation Location	USB Image	Hash Status
KittyMontage.mov	Desktop\Pics\Hidden\Videos\	jo-favorites-usb	Identical MD5 ✓
Cat.mov	Desktop\Pics\Hidden\Videos\	jo-work-usb	Identical MD5 ✓

MontereyKitty.m4v	Desktop\Pics\Hidden\Videos\	jo-favorites-usb	Identical MD5 ✓
MontereyKittyHQ.m4v	Desktop\Pics\Hidden\Videos\	jo-favorites-usb	Identical MD5 ✓
TiggerTheCat.m4v	Desktop\Pics\Hidden\Videos\	jo-favorites-usb	Identical MD5 ✓

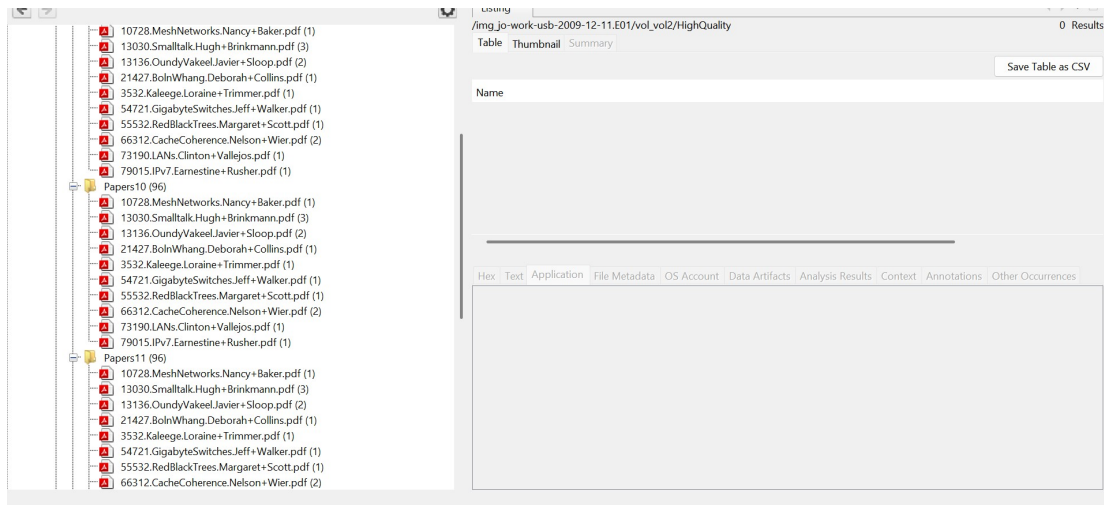


Figure 8 — File duplication confirmed across USB devices (hash match)

6. Correlated Behavioural Timeline

Timestamp (UTC)	Event Type	Artefact	Forensic Significance
2009-11-23 13:23:53	Software Install	Registry	Python 2.6.4 installed — runtime for patentauto.py
2009-11-23 18:20:12	Download	Firefox	python-2.6.4.msi downloaded from python.org
2009-11-23 18:24:50	File Access	LNK	DIIT steganography tool (diit-1.5.jar) opened
2009-11-23 18:24:51	Folder Browse	LNK	Pics folder accessed — steganography workspace
2009-11-23 18:25:47	File Created	Metadata	testHide.bmp created
2009-11-23 18:25:53	File Access	LNK	testHide.bmp accessed — 63 s after DIIT
2009-11-23 21:54:44	USB Connected	SYSTEM hive	Alcor Micro Flash Drive attached
2009-11-23 21:55:01	File Created	Metadata	patentauto.py created
2009-11-23 21:55:45	File Access	LNK	patentauto.py accessed — 61 s after USB attach
2009-11-23 21:55:46	Folder Browse	LNK	web/ folder accessed
2009-11-23 21:57:24	File Modified	LNK	patentauto.lnk last modified
2009-11-24 22:01:49	USB Connected	SYSTEM hive	Unknown USB (Vid_0000&Pid_7777) attached
2009-11-24 22:01:53	File Access	LNK	STU.doc accessed — 4 s after USB attach
2009-11-24 22:08:36	Download	Firefox	QuickTimeInstaller.exe downloaded
2009-11-24 22:14:30	File Access	LNK	DSC00065 / DSC00070 high-res images viewed

A	B	C	D	E	F	G	H
Timestamp	Artifact	Path	Significance				
23/11/2009 18:25	testHide.bmp created	Desktop/Pics	hidden-image investigation				
23/11/2009 18:25	testHide.lnk created	Recent folder	confirms access				
23/11/2009 21:55	patentauto.py created	Desktop/web	automation script				
23/11/2009 21:55	patentauto.lnk created	Recent folder	execution evidence				
23/11/2009 21:57	patentauto.lnk modified	Recent folder	active usage				
16/11/2009 0:22	mdd_1.3.exe executed	RAM acquisition	memory capture				

Figure 9 — Timeline view in Autopsy

The screenshot shows the Autopsy interface with a keyword search for 'testhide' yielding 22 results. The 'Table' tab is selected, showing a list of artifacts. The 'TL Events' tab is also visible, showing a detailed view of a specific event.

Name	Keyword Preview	Location	Modified Time
change.log.1	settings\jo\recent\testhide.lnk\documents and sett	/img_jo-2009-11-24.E01/vol_vol2/System Volume In...	2009-11-24
TL Events Artifact	ngs\jo\desktop\pics\testhide.bmp> relative path: \d	/img_jo-2009-11-24.E01/vol_vol2/Documents and S...	2009-11-23
_REGISTRY_USER_NTUSER_S-1-5-21-606747145-15	x\oldtesthide.bmp<testhide.lnk<testhide.lnk<test	/img_jo-2009-11-24.E01/vol_vol2/System Volume In...	2009-11-24
\$UsnJmL\$J	.bmp<testhide.bmp<<testhide.lnk<<testhide.lnk<test	/img_jo-2009-11-24.E01/vol_vol2/\$Extend/\$UsnJmL\$J	2009-11-20
TL Events Artifact	ngs\jo\desktop\pics\testhide.bmp> relative path: \d	/img_jo-2009-11-24.E01/vol_vol2/Documents and S...	2009-11-23
TL Events Artifact	ngs\jo\desktop\pics\testhide.bmp> relative path: \d	/img_jo-2009-11-24.E01/vol_vol2/Documents and S...	2009-11-23
LogFile	jn r-<jltest.png>testhide.bmp>thumbs.db\$301\$3	/img_jo-2009-11-24.E01/vol_vol2/LogFile	2009-11-20
SMFT	o.lnkpics.lnk>testhide.lnk>file\printh-1dpri	/img_jo-2009-11-24.E01/vol_vol2/SMFT	2009-11-20

Type	Value	Source(s)
Date/Time	2009-11-23 18:25:47 UTC	Plaso
Description	[Empty description] File size: 9054 File attribute flags: 0x00000020 Drive type: 3 Drive serial number: 0x582a0109 Local path: C:\Documents and Settings\Jo\Desktop\Pics\testhide.bmp Relative path: \Desktop\Pics\testhide.bmp Working dir: C:\Documents and Settings\Jo\Desktop\Pics\Li nk target: Pics\testhide.bmp	Plaso
Event Type	23	Plaso
Source File Path	/img_jo-2009-11-24.E01/vol_vol2/Documents and Settings\Jo\Recent\testHide.lnk	
Artifact ID	-9223372036854769062	

Figure 10 — Timeline event-to-artifact correlation